

exercise screenshots showing code works:

1.

```
brendanbaron@MacBook-Air A3 % python3 E1rep-server.py
Received request: b'Hello, server!'
Received request: b'Hello, server!'
Received request: b'Hello, server!'
[Ctrl-C]
```

```
[brendanbaron@MacBook-Air A3 % python3 E1req-client.py
Sending request Client 1 ...
Received reply 1 [ b'Hello, client!' ]
Sending request Client 2 ...
Received reply 2 [ b'Hello, client!' ]
Sending request Client 3 ...
```

2.

```
Received request: b'Hello from client 1'
Received request: b'Hello from client 2'
Received request: b'Hello from client 3'
Received request: b'Hello from client 1'
Received request: b'Hello from client 2'
Received request: b'Hello from client 3'
```

(there's one of each of these for each of the 3 clients)

```
brendanbaron@MacBook-Air A3 % python3 E2req-client1.py
Sending request Client 1 1 ...
Received reply 1 [ b'Hello, client!' ]
Sending request Client 1 2 ...
Received reply 2 [ b'Hello, client!' ]
Sending request Client 1 3 ...
Received reply 3 [ b'Hello, client!' ]
Sending request Client 1 4 ...
Received reply 4 [ b'Hello, client!' ]
Sending request Client 1 5 ...
Received reply 5 [ b'Hello, client!' ]
Sending request Client 1 6 ...
Received reply 6 [ b'Hello, client!' ]
Sending request Client 1 7 ...
Received reply 7 [ b'Hello, client!' ]
Sending request Client 1 8 ...
Received reply 8 [ b'Hello, client!' ]
Sending request Client 1 9 ...
Received reply 9 [ b'Hello, client!' ]
brendanbaron@MacBook-Air A3 %
```

3.

```
brendanbaron@MacBook-Air A3 % python3 E3pub-server.py 127.0.0.1 5560

Publishing: News update: something happened
Publishing: News update: something happened
Publishing: News update: something happened

brendanbaron@MacBook-Air A3 % python3 E3pub-client.py 127.0.0.1 5560
Received: News update: something happened
Received: News update: something happened
Received: News update: something happened
```

4.

```
brendanbaron@MacBook-Air A3 % python3 E4ventilator.py
Sending: Task 0
Sending: Task 1
Sending: Task 2
Sending: Task 3
Sending: Task 4
Sending: Task 5
Sending: Task 6
Sending: Task 7
Sending: Task 8
Sending: Task 9
```

```
brendanbaron@MacBook-Air A3 % python3 E4worker1.py
Worker 1 processing Task 2
Worker 1 processing Task 4
Worker 1 processing Task 6
Worker 1 processing Task 8
```

```
brendanbaron@MacBook-Air A3 % python3 E4worker2.py
Worker 2 processing Task 0
Worker 2 processing Task 1
Worker 2 processing Task 3
Worker 2 processing Task 5
Worker 2 processing Task 7
Worker 2 processing Task 9
```

5.

```
brendanbaron@MacBook-Air A3 % python3 E5dealer1.py
Client client_664 connected
Sent message: Hello from client_664
Received message: Hello from client_534
Sent message: Hello from client_664
Received message: Hello from client_534
Sent message: Hello from client_664
Received message: Hello from client_534
```

```
[brendanbaron@MacBook-Air A3 % python3 E5dealer2.py
Client client_534 connected
Sent message: Hello from client_534
Received message: Hello from client_664
Sent message: Hello from client_534
Received message: Hello from client_664
```

6.

```
(pyro) brendanbaron@MacBook-Air src % python3 E6server.py
Ready. Object uri = PYRO:obj_c2e036ab58064b779d3e3066daf8cc65@localhost:65120
greet() called with name: Brendan

(pyro) brendanbaron@MacBook-Air src % python3 E6client.py
Enter the Pyro URI of the Greeter class: PYRO:obj_c2e036ab58064b779d3e3066daf8cc65@localhost:65120
What is your name? Brendan
Hello, Brendan!
```

another version i made:

```
(pyro) brendanbaron@MacBook-Air src % python3 E6client.py
Enter the Pyro URI of the Greeter class: PYRO:obj_2a1734d7cd4f4dada5bf29075e3d8316@localhost:65133

Choose an action: 'greet', 'history', or 'quit': greet
Enter a name to greet: Brendan
Hello, Brendan!

Choose an action: 'greet', 'history', or 'quit': greet
Enter a name to greet: Stelios
Hello, Stelios!

Choose an action: 'greet', 'history', or 'quit': history
Names greeted so far: ['Brendan', 'Stelios']

Choose an action: 'greet', 'history', or 'quit': quit
Goodbye!
```

7.


```
(pyro) brendanbaron@MacBook-Air src % python3 E7server.py
Ready. Object uri = PYRO:obj_4216447b26a8477c9611578940ae70e2@localhost:51944
add(4, 3) called
subtract(4, 3) called
divide(8, 2) called
```

```
Enter the Pyro URI of the ArithmeticServer: PYRO:obj_4216447b26a8477c9611578940ae70e2@localhost:51944
```

Choose an operation:

1. Add
2. Subtract
3. Divide
4. Quit

Enter your choice (1-4): 1

Enter the first number: 4

Enter the second number: 3

Result: 7.0

Choose an operation:

1. Add
2. Subtract
3. Divide
4. Quit

Enter your choice (1-4): 2

Enter the first number: 4

Enter the second number: 3

Result: 1.0

Choose an operation:

1. Add
2. Subtract
3. Divide
4. Quit

Enter your choice (1-4): 3

Enter the first number: 8

Enter the second number: 2

Result: 4.0

Choose an operation:

1. Add
2. Subtract
3. Divide
4. Quit

Enter your choice (1-4): 4

Goodbye!

```
(pyro) brendanbaron@MacBook-Air src % python3 E8server.py
Ready. Object uri = PYRO:obj_1a3b7cde657c423bb3aed4c629253dbd@localhost:52008
[2025-04-13 14:41:48] Client 0103 called reverse_list
[2025-04-13 14:42:03] Client 0103 called remove_duplicates
```

```
(pyro) brendanbaron@MacBook-Air src % python3 E8client.py
Enter the Pyro URI of the ListProcessor: PYRO:obj_1a3b7cde657c423bb3aed4c629253dbd@localhost:52008
Enter your unique client ID: 0103
```

Choose a list operation:

1. Reverse List
2. Remove Duplicates
3. Quit

Enter choice (1-3): 1

Enter a comma-separated list (e.g. a,b,b,c): b,r,e,n,d,a,n

Reversed List: ['n', 'a', 'd', 'n', 'e', 'r', 'b']

Choose a list operation:

1. Reverse List
2. Remove Duplicates
3. Quit

Enter choice (1-3): 2

Enter a comma-separated list (e.g. a,b,b,c): b,r,e,n,d,a,n

List without duplicates: ['b', 'r', 'e', 'n', 'd', 'a']

Choose a list operation:

1. Reverse List
2. Remove Duplicates
3. Quit

Enter choice (1-3): 3

Goodbye!

```
(pyro) brendanbaron@MacBook-Air src % python3 E9server.py
Ready. Object uri = PYRO:obj_96f481f2ea8048abb5d9e8e3efe9078c@localhost:52040
word_count called: 17 words
most common word called: 'doing' (2 times)
```

```
[(pyro) brendanbaron@MacBook-Air src % python3 E9client.py
Enter the Pyro URI of the TextAnalyzer: PYRO:obj_96f481f2ea8048abb5d9e8e3efe9078c@localhost:52040
```

Choose an analysis:

1. Word Count
2. Most Common Word
3. Quit

Enter your choice (1-3): 1

Choose input method:

1. Type text manually
2. Load text from a file

Enter 1 or 2: 1

Enter your text: hi how are you doing im going to rome next week. i am doing web dev assessment

Word Count: 17

Choose an analysis:

1. Word Count
2. Most Common Word
3. Quit

Enter your choice (1-3): 2

Choose input method:

1. Type text manually
2. Load text from a file

Enter 1 or 2: 2

Enter the file path: E9text.txt

Most Common Word: doing

Choose an analysis:

1. Word Count
2. Most Common Word
3. Quit

Enter your choice (1-3): 3

Goodbye!

```
[(pyro) brendanbaron@MacBook-Air src % python3 E10load-balancer.py  
LoadBalancer is live.
```

```
LoadBalancer: Forwarded mean to stats1
```

```
LoadBalancer: Forwarded median to stats2
```

```
LoadBalancer: Forwarded median to stats1
```

```
[(pyro) brendanbaron@MacBook-Air src % python3 E10client.py
```

```
Options:
```

- 1. Calculate Mean
- 2. Calculate Median
- 3. Quit

```
Choose (1-3): 1
```

```
Enter comma-separated numbers (e.g., 1,2,3): 2,5,4,67,23,11
```

```
Result: 18.666666666666668 (from stats1)
```

```
Options:
```

- 1. Calculate Mean
- 2. Calculate Median
- 3. Quit

```
Choose (1-3): 2
```

```
Enter comma-separated numbers (e.g., 1,2,3): 3,5,7,99,100,2,66
```

```
Result: 7.0 (from stats2)
```

```
Options:
```

- 1. Calculate Mean
- 2. Calculate Median
- 3. Quit

```
Choose (1-3): 2
```

```
Enter comma-separated numbers (e.g., 1,2,3): 1,1,1,1
```

```
Result: 1.0 (from stats1)
```

```
Options:
```

- 1. Calculate Mean
- 2. Calculate Median
- 3. Quit

```
Choose (1-3): 3
```